Migration Choice In Levels. These were simply

$$\pi^n_{ll',t} = \frac{\exp\left(\beta V^n_{l',t+1} - (\kappa_{ll'} + m_{ll',t})\right)^{\frac{1}{\nu^n}}}{\sum_{k\in\mathcal{L}} \exp\left(\beta V^n_{k,t+1} - (\kappa_{ll'} + m_{ll',t})\right)^{\frac{1}{\nu^n}}}$$

Let $u^n_{l,t} = \exp V^n_{l,t}$ and $\mu_{ll',t} = \exp m_{ll',t}$. The equilibrium system of equations we can use to solve the model without having to estimate bilateral flows comes from writing values and migration rates in changes.

Migration Rates in Changes

$$\begin{aligned}
\frac{\pi^n_{ll',t}}{\pi^n_{ll',t-1}} &= \frac{\exp(\beta V^n_{l',t+1} - (\kappa_{ll'} + m_{ll',t}))^{\frac{1}{\nu^n}}}{\exp(\beta V^n_{l',t} - (\kappa_{ll'} + m_{ll',t-1}))^{\frac{1}{\nu^n}}} \times \frac{\sum_{k\in\mathcal{L}} \exp\left(\beta V^n_{k,t} - (\kappa_{lk} + m_{lk,t-1})\right)^{\frac{1}{\nu^n}}}{\sum_{j\in\mathcal{L}} \exp\left(\beta V^n_{j,t+1} - (\kappa_{lj} + m_{lj,t})\right)^{\frac{1}{\nu^n}}} \\
&= \frac{(\dot{u}^n_{l',t+1})^{\frac{\beta}{\nu^n}} (\dot{\mu}_{ll',t})^{\frac{1}{\nu^n}}}{\sum_j \frac{\exp(\beta V^n_{j,t+1} - (\kappa_{lj} + m_{lj,t}))^{\frac{1}{\nu^n}}}{\sum_k \exp(\beta V^n_{k,t} - (\kappa_{lk} + m_{lk,t-1}))^{\frac{1}{\nu^n}}}} \\
\iff \pi^n_{ll',t} &= \frac{(\dot{u}^n_{l',t+1})^{\frac{\beta}{\nu^n}} (\dot{\mu}_{ll',t})^{\frac{1}{\nu^n}}}{\sum_j \pi^n_{lj,t-1} (\dot{u}^n_{j,t+1})^{\frac{\beta}{\nu^n}} (\dot{\mu}_{lj,t})^{\frac{1}{\nu^n}}} \pi^n_{ll',t-1}
\end{aligned}$$

EMAX in Changes.

Start by exponentiating the EMAX operator

$$\begin{aligned}
u^n_{l,t} &= w_{l,t} \left(\sum_{l'\in\mathcal{L}} \exp\left(\beta V^n_{l',t+1} - (\kappa_{l'l} + m_{l'l,t})\right)^{\frac{1}{\nu^n}} \right)^{\nu_n} \\
\iff \dot{u}^n_{l,t} &= \dot{w}_{l,t} \left(\sum_{l'\in\mathcal{L}} \pi^n_{ll',t-1} (\dot{u}^n_{l',t+1})^{\frac{\beta}{\nu_n}} (\dot{\mu}_{ll',t})^{\frac{1}{\nu_n}} \right)^{\nu_n}
\end{aligned}$$

Aggregate Labor Supply.

$$\begin{aligned}
L_{l,t} &= \left[\lambda^{1-\rho}_{l,t} (\alpha^F L^F_{l,t})^\rho + (1-\lambda_{l,t})^{1-\rho} (\alpha^D L^D_{l,t})^\rho \right]^{\frac{1}{\rho}} \\
&= \left[\lambda^{1-\rho}_{l,t} (\alpha^F \dot{L}^F_{l,t})^\rho (L^F_{l,t-1})^\rho + (1-\lambda_{l,t})^{1-\rho} (\alpha^D \dot{L}^D_{l,t})^\rho (L^D_{l,t-1})^\rho \right]^{\frac{1}{\rho}}
\end{aligned}$$

Aggregate Productivity.

Appendix of the paper has the derivation of Z with $z^D(\tau) = e^{\Gamma + \gamma^D \tau}$ and $z^F(\tau) = e^{\gamma^F \tau}$. This is equation A.11 from the paper

$$
\begin{aligned}
Z_{s,t}^{\frac{\rho}{1-\rho}} &= \int_{\Omega}\left[\int_0^{\mathcal{T}_{s,t}(\omega)} (\omega z^F(\tau))^{\frac{\rho}{1-\rho}} d\tau + \int_{\mathcal{T}_{s,t}(\omega)}^1 z^D(\tau)^{\frac{\rho}{1-\rho}} d\tau\right] dG(\omega) \\
&= \left(\frac{1-\rho}{\gamma^F \rho}\right)\left[\left(\frac{\alpha^F w_{s,t}^D}{\alpha^D w_{s,t}^F} \times \frac{1}{e^\Gamma}\right)^{\frac{\rho\gamma^F}{(1-\rho)(\gamma^D-\gamma^F)}} \mathbb{E}\omega^{\frac{\rho\gamma^D}{(1-\rho)(\gamma^D-\gamma^F)}} - \mathbb{E}\omega^{\frac{\rho}{1-\rho}}\right] \\
&+ e^{\frac{\rho}{1-\rho}\Gamma}\frac{1-\rho}{\rho\gamma^D}\left[e^{\frac{\rho\gamma^D}{1-\rho}} - \left(\frac{\alpha^F w_{s,t}^D}{\alpha^D w_{s,t}^F} \times \frac{1}{e^\Gamma}\right)^{\frac{\rho\gamma^D}{(1-\rho)(\gamma^D-\gamma^F)}} \mathbb{E}\omega^{\frac{\rho\gamma^D}{(1-\rho)(\gamma^D-\gamma^F)}}\right] \qquad \text{(A.11)}
\end{aligned}
$$

Resource Feasibility. Capital demand comes from standard Cobb-Douglas demand

$$
\begin{aligned}
(r+\delta)K_{l,t} &= \theta Y_{l,t} = \theta K_{l,t}^\theta (Z_{l,t}L_{l,t})^{1-\theta} \\
K_{l,t} &= \left(\frac{\theta}{r+\delta}\right)^{\frac{1}{1-\theta}} Z_{l,t}L_{l,t}
\end{aligned}
$$

where r, δ are exogenous. Then,

$$
\begin{aligned}
Y_{l,t} &= w_{l,t}^D L_{l,t}^D + w_{l,t}^F L_{l,t}^F + (r+\delta)\left(\frac{\theta}{r+\delta}\right)^{\frac{1}{1-\theta}} Z_{l,t}L_{l,t} \\
K_{l,t}^\theta (Z_{l,t}L_{l,t})^{1-\theta} &= w_{l,t}^D L_{l,t}^D + w_{l,t}^F L_{l,t}^F + (r+\delta)\left(\frac{\theta}{r+\delta}\right)^{\frac{1}{1-\theta}} Z_{l,t}L_{l,t} \\
\left(\frac{\theta}{r+\delta}\right)^{\frac{\theta}{1-\theta}} Z_{l,t}L_{l,t} &= w_{l,t}^D L_{l,t}^D + w_{l,t}^F L_{l,t}^F + (r+\delta)\left(\frac{\theta}{r+\delta}\right)^{\frac{1}{1-\theta}} Z_{l,t}L_{l,t} \\
\left(\frac{\theta}{r+\delta}\right)^{\frac{\theta}{1-\theta}} Z_{l,t}L_{l,t}\left[1-(r+\delta)\left(\frac{\theta}{r+\delta}\right)\right] &= w_{l,t}^D L_{l,t}^D + w_{l,t}^F L_{l,t}^F \\
\left(\frac{\theta}{r+\delta}\right)^{\frac{\theta}{1-\theta}} (1-\theta) Z_{l,t}L_{l,t} &= w_{l,t}^D L_{l,t}^D + w_{l,t}^F L_{l,t}^F
\end{aligned}
$$

Solving the Model

The algorithm for solving follows this follows.

0. Initialize

- Initialize some pre-period distribution of migration probabilities across space, $\pi_{ll',-1}^n$. When I turn to estimation, this will be taken from the data. For now just set them equal.
- Initialize time-zero stocks of labor across regions. Again, just go with uniform distribution.

1. Solve for Wages Across Locations

I.e. solve the static problem.

We have three equations in each location for this. The paper also introduces some functional forms. Let $z^D(\tau) = e^{\Gamma+\gamma^D\tau}$ and $z^F(\tau) = e^{\gamma^F\tau}$ where $\gamma^D > \gamma^F$ by comparative advantage.

Combined with the Pareto distributed ω assumption this yields (appendix equation A.11)

$$
\begin{aligned}
Z_{l,t}^{\frac{\rho}{1-\rho}} &= \int_\Omega \left[\int_0^{\mathcal{T}_{l,t}(\omega)} (\omega z^F(\tau))^{\frac{\rho}{1-\rho}} d\tau + \int_{\mathcal{T}_{l,t}(\omega)}^1 z^D(\tau)^{\frac{\rho}{1-\rho}} d\tau \right] dG(\omega) \\
&= \left(\frac{1-\rho}{\gamma^F \rho}\right) \left[\left(\frac{\alpha^F w_{l,t}^D}{\alpha^D w_{l,t}^F} \times \frac{1}{e^\Gamma} \right)^{\frac{\rho\gamma^F}{(1-\rho)(\gamma^D-\gamma^F)}} \mathbb{E}\omega^{\frac{\rho\gamma^D}{(1-\rho)(\gamma^D-\gamma^F)}} - \mathbb{E}\omega^{\frac{\rho}{1-\rho}} \right] \\
&+ e^{\frac{\rho}{1-\rho}\Gamma} \frac{1-\rho}{\rho\gamma^D} \left[e^{\frac{\rho\gamma^D}{1-\rho}} - \left(\frac{\alpha^F w_{l,t}^D}{\alpha^D w_{l,t}^F} \times \frac{1}{e^\Gamma} \right)^{\frac{\rho\gamma^D}{(1-\rho)(\gamma^D-\gamma^F)}} \mathbb{E}\omega^{\frac{\rho\gamma^D}{(1-\rho)(\gamma^D-\gamma^F)}} \right]
\end{aligned}
$$

where we can see that

$$
\int_\Omega \left[\int_0^{\mathcal{T}_{l,t}(\omega)} \omega z^F(\tau)^{\frac{\rho}{1-\rho}} d\tau \right] dG(\omega) := \left(\frac{1-\rho}{\gamma^F \rho}\right) \left[\left(\frac{\alpha^F w_{l,t}^D}{\alpha^D w_{l,t}^F} \times \frac{1}{e^\Gamma} \right)^{\frac{\rho\gamma^F}{(1-\rho)(\gamma^D-\gamma^F)}} \mathbb{E}\omega^{\frac{\rho\gamma^D}{(1-\rho)(\gamma^D-\gamma^F)}} - \mathbb{E}\omega^{\frac{\rho}{1-\rho}} \right]
$$

and

$$
\int_\Omega \left[\int_{\mathcal{T}_{l,t}(\omega)}^1 z^D(\tau)^{\frac{\rho}{1-\rho}} d\tau \right] dG(\omega) := e^{\frac{\rho}{1-\rho}\Gamma} \frac{1-\rho}{\rho\gamma^D} \left[e^{\frac{\rho\gamma^D}{1-\rho}} - \left(\frac{\alpha^F w_{l,t}^D}{\alpha^D w_{l,t}^F} \times \frac{1}{e^\Gamma} \right)^{\frac{\rho\gamma^D}{(1-\rho)(\gamma^D-\gamma^F)}} \mathbb{E}\omega^{\frac{\rho\gamma^D}{(1-\rho)(\gamma^D-\gamma^F)}} \right].
$$

Also, the appendix of the paper calculates the two expectations involved

$$
\mathbb{E}\omega^{\frac{\rho}{1-\rho}\frac{\gamma^D}{\gamma^D-\gamma^F}} = \frac{(\gamma^D-\gamma^F)(1-\rho)}{(\gamma^D-\gamma^F)(1-\rho)-\rho\gamma^D} \quad \text{and} \quad \mathbb{E}\omega^{\frac{\rho}{1-\rho}} = \frac{1-\rho}{1-2\rho}
$$

These two integrals are important because they show up directly in the labor market clearing conditions, which, together with the resource feasibility condition pin down wages and task prices

$$
\begin{aligned}
\left(\frac{w_{l,t}^D}{P_{l,t}}\right)^{-\frac{\rho}{1-\rho}} &= \left(\frac{L_{l,t}^D}{Q_{l,t}}\right)^\rho \left(\int_\Omega \int_{\mathcal{T}_{l,t}(\omega)}^1 (\alpha^D z^D(\tau))^{\frac{\rho}{1-\rho}} d\tau dG(\omega) \right)^{-\rho} \\
\left(\frac{w_{l,t}^F}{P_{l,t}}\right)^{-\frac{\rho}{1-\rho}} &= \left(\frac{L_{l,t}^F}{Q_{l,t}}\right)^\rho \left(\int_\Omega \int_0^{\mathcal{T}_{l,t}(\omega)} (\omega\alpha^F z^F(\tau))^{\frac{\rho}{1-\rho}} d\tau dG(\omega) \right)^{-\rho} \\
\left(\frac{\theta}{r+\delta}\right)^{\frac{\theta}{1-\theta}} (1-\theta) Z_{l,t} L_{l,t} &= w_{l,t}^D L_{l,t}^D + w_{l,t}^F L_{l,t}^F
\end{aligned}
$$

where recall that

- $Z_{l,t} = \left(\int_\Omega \left[\int_0^{\mathcal{T}_\omega} (\omega z^F(\tau))^{\frac{\rho}{1-\rho}} d\tau + \int_{\mathcal{T}_\omega}^1 z^D(\tau)^{\frac{\rho}{1-\rho}} d\tau \right] dG(\omega) \right)^{\frac{1-\rho}{\rho}}$
- $L_{l,t} = \left[\lambda_{l,t}{}^{1-\rho} (\alpha^F L_{l,t}^F)^\rho + (1-\lambda_{l,t})^{1-\rho} (\alpha^D L_{l,t}^D)^\rho \right]^{\frac{1}{\rho}}$
- $\lambda_{l,t} = \int_\Omega \left[\int_0^{\mathcal{T}_\omega} (\omega z^F(\tau))^{\frac{\rho}{1-\rho}} d\tau \right] dG(\omega) / Z_{l,t}^{\frac{\rho}{1-\rho}}$

Note. $Q_{l,t} = Z_{l,t} L_{l,t}$ in equilibrium which should be in the code since the integrals also depend on the relative wages (i.e. don't simply solve over Q)

2. Update the Pi's

To do this let Π_l be the T vector of migration rates for location l from time 0 to $T-1$.

$$\pi^n_{ll',t} = \frac{(\dot{u}^n_{l',t+1})^{\frac{\beta}{\nu^n}} (\dot{\mu}_{ll',t})^{\frac{1}{\nu^n}}}{\sum_j \pi^n_{lj,t-1} (\dot{u}^n_{j,t+1})^{\frac{\beta}{\nu^n}} (\dot{\mu}_{lj,t})^{\frac{1}{\nu^n}}} \pi^n_{ll',t-1}$$

Remark. Julia is indexed starting at 1 so it is best in the code to think of the initial period as one not zero. Then, to calculate the updated elements of Π_l requires,

1. $[u_2/u_1, \ldots, u_T/u_{T-1}]$ which is the $T-1$ length vector of value changes.
2. But these $\dot{u}$ must be updated before calculating the update of Π_l using

$$\dot{u}^n_{l,t} = \dot{w}^n_{l,t} \left(\sum_{l' \in \mathcal{L}} \pi^n_{ll',t-1} (\dot{u}^n_{l',t+1})^{\frac{\beta}{\nu_n}} (\dot{\mu}_{ll',t})^{\frac{1}{\nu_n}} \right)^{\nu_n} \$\$ **Remark.** Calculating \$\dot{u}_T\$ from this requires the$$

3. Update the L's using the Law of Motion

$$L^n_{l,t+1} = \sum_{l' \in \mathcal{L}} \min \left\{ \pi^n_{l'l,t} L^n_{l',t}, M^n_{l'l} \right\}$$

where M is a quota on flows from l' to l.

4. Calculate the Error Between New and Previous $\dot{u}$

Return to step 1 if above tolerance, otherwise terminate loop.